Department of Education

SHS-TVL

Oracle Database Programming NC III

Module 2

RYAN JEFF MERCADO
Writer

Melanio San Jose
Validator

What I Need to Know

A database management system (DBMS) refers to the technology for creating and managing databases. DBMS is a software tool to organize (create, retrieve, update, and manage) data in a database.

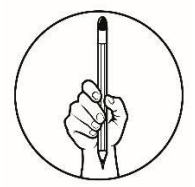
The proper understanding of data structures and algorithms will help you to understand the DBMS quickly.

The module has the following:

- Lesson 1 – Database Essentials
    - Database Environment
    - Database Architecture
    - Relation Data Model
    - DBMS Data Models
    - DBMS Data Independence
    - Entity Relationship (ER) Model

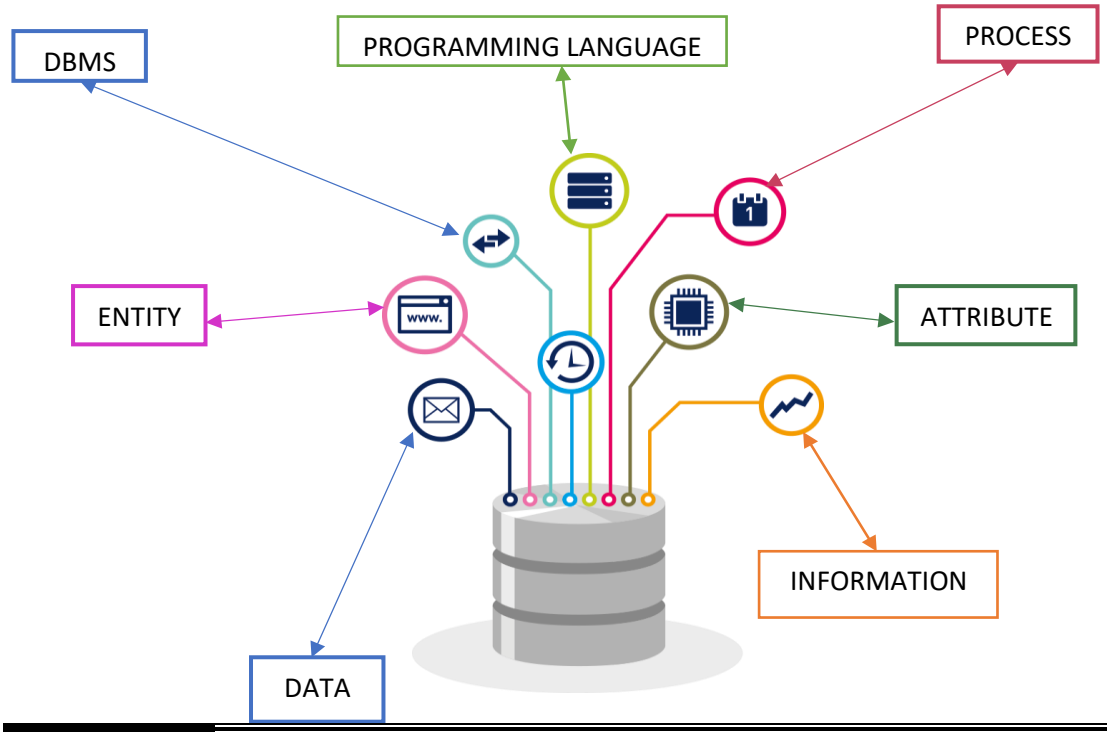
After going through this module, you are expected to:

1. Define basic terminologies in Database Environment.
2. Enumerate the different types of databases.
3. Define and enumerate the DBMS architecture.
4. Define Relational Model.
5. Define and Enumerate Database Schema.Define Data Independence.
6. Define the use of ER Model.
7. Create a simple database system using the ER Model.



What I Know

As defined, database management system is the technology to create and manage databases. Now, what do you think make-up a good DBMS? A Mind-Map is given below and all you need to do is to distinguish which are the terms that builds up a good DBMS. Upon selection, try to define the terms in your own words and carry your knowledge all throughout the module.



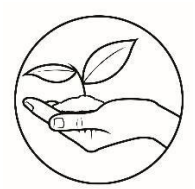
Lesson 1 Database Management System Essentials

A database management system (DBMS) refers to the technology for creating and managing databases. DBMS is a software tool to organize (create, retrieve, update, and manage) data in a database.

The main aim of a DBMS is to supply a way to store up and retrieve database information that is both convenient and efficient. By data, we mean known facts that can be recorded and that have embedded meaning. Usually, people use software such as DBASE IV or V, Microsoft ACCESS, or EXCEL to store data in the form of a database. A datum is a unit of data. Meaningful data combined to form information. Hence, information is interpreted data - data provided with semantics. MS. ACCESS is one of the most common examples of database management software.

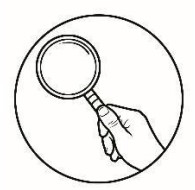
What's In

A pre-requisite knowledge before this module are the following from Module 1: Computer Programming – Overview, Environment and Basics and Basic Syntax.



What's New

From Module 1: Computer Programming Essentials, we learned different programming languages, there similarities and differences in terms of algorithm. On this lesson, we will focus on one Programming Language – Structured Query Language (SQL) and Database Management System (DBMS) Essentials.



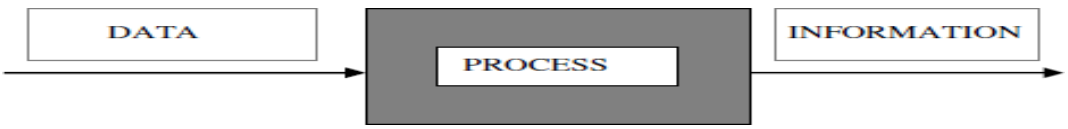
What is It

I: OVERVIEW

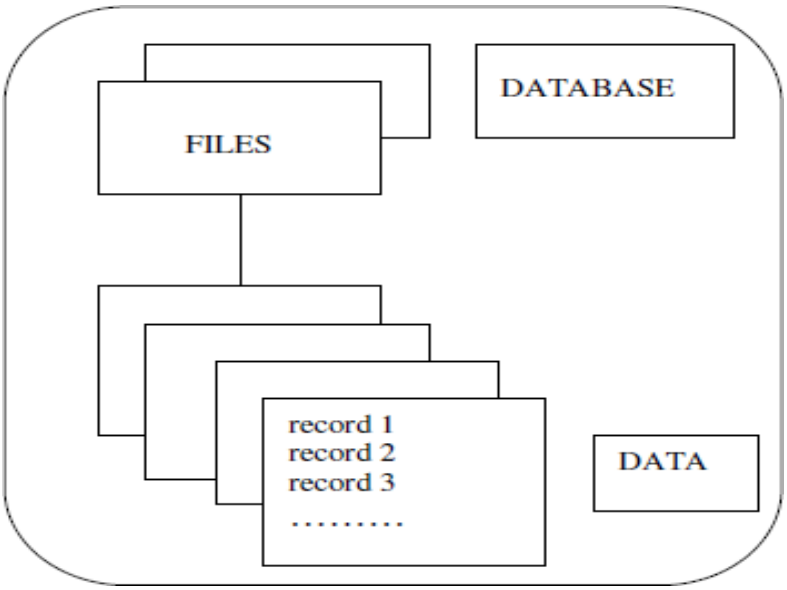
A **DATABASE** is one of the essential components for many applications and is used for storing a series of data in a single set. In other words, it is a group/package of information that is put in order so that it can be easily accessed, manage, and update.

There are different types of databases. They are:

- Bibliographic
- full-text
- numeric
- images



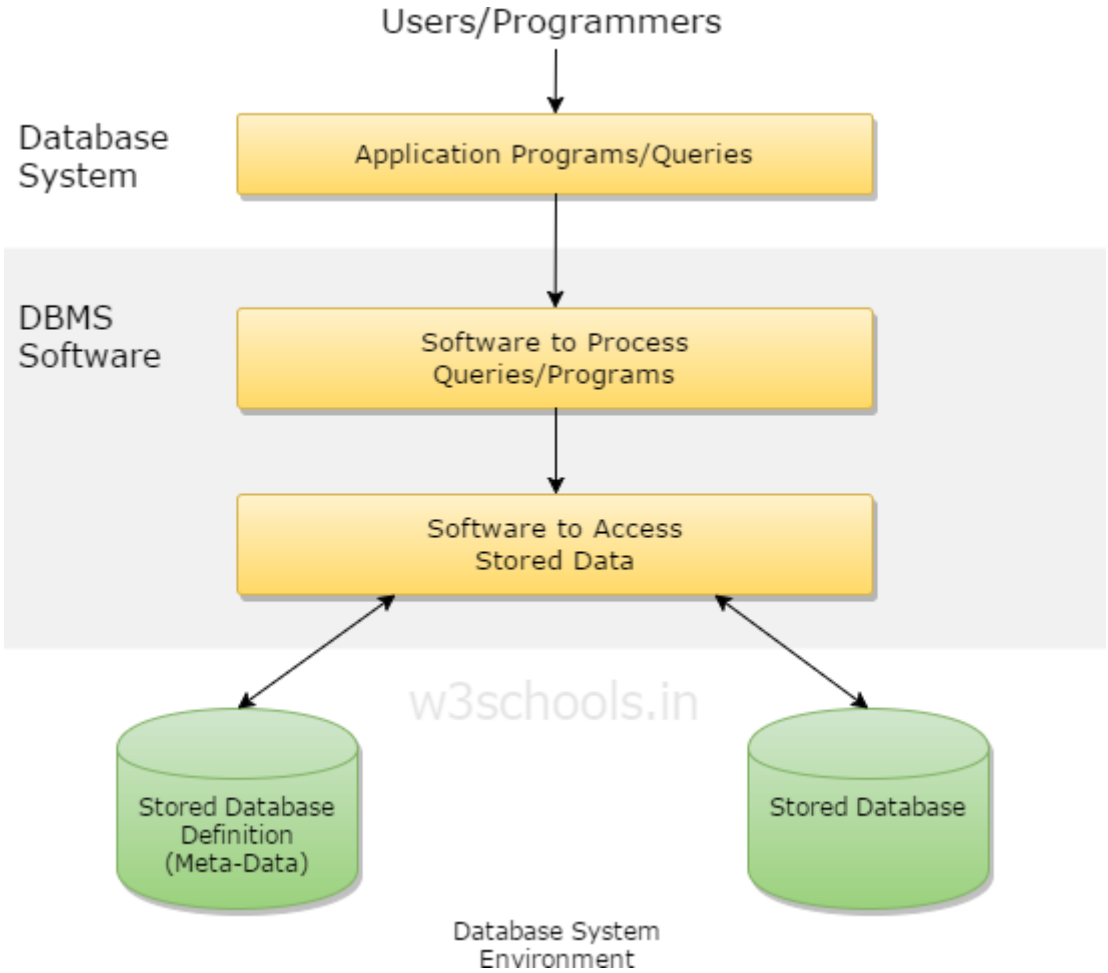
- **Data** are facts concerning things such as people, objects or events.
- **Information** is data that have been processed and presented in a form suitable for human interpretation, often with the purpose of revealing trends or patterns.
- **Process** involves acquisition, storage, manipulation, retrieval and distribution.
- **Database** is a shared collection of logically related data, designed to meet the information needs of multiple users in an organization.



II: DATABASE ENVIRONMENT

A database environment is a collective system of components that comprise and regulates the group of data, management, and use of data, which consist of software, hardware, people, techniques of handling database, and the data also.

Here, the hardware in a database environment means the computers and computer peripherals that are being used to manage a database, and the software means the whole thing right from the operating system (OS) to the application programs that include database management software like M.S. Access or SQL Server. Again, the people in a database environment include those people who administrate and use the system. The techniques are the rules, concepts, and instructions given to both the people and the software along with the data with the group of facts and information positioned within the database environment.



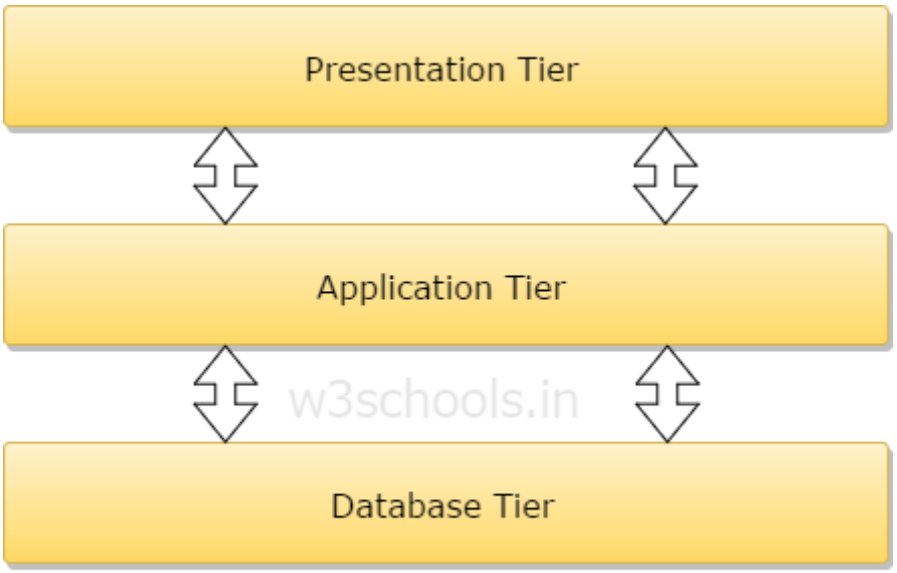
III: DATABASE ARCHITECTURE

A DBMS architecture is depending on its design and can be of the following types:

- Centralized
- Decentralized
- Hierarchical

DBMS architecture can be seen as either a single-tier or multi-tier. An architecture having n-tier splits the entire system into related but independent n modules that can be independently customized, changed, altered, or replaced.

The architecture of a database system is very much influenced by the primary computer system on which the database system runs. Database systems can be centralized, or client-server, where one server machine executes work on behalf of multiple client machines. Database systems can also be designed to exploit parallel computer architectures. Distributed databases span multiple geographically separated machines.



The Three Tier Architecture

IV: RELATIONAL MODEL

The relational model is the theoretical basis of relational databases, which is a technique or way of structuring data using relations, which are grid-like mathematical structures consisting of columns and rows. Codd proposed the relational model for IBM, but the idea became extremely vital and prominent that his work would become the basis of relational databases. You might be very familiar with the physical demonstration of a relation in a database - which is known as a *table*.

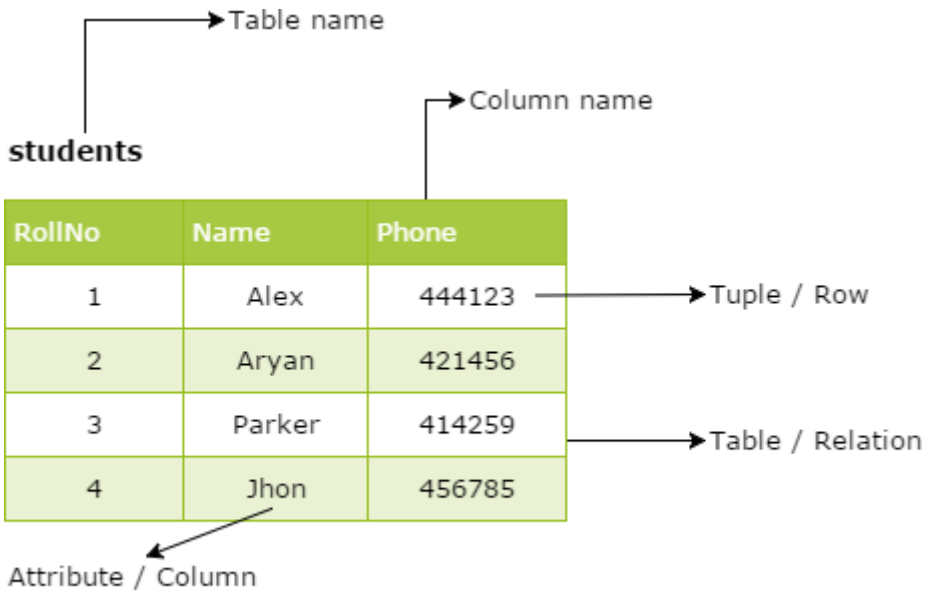
In the relational model, all data is logically structured within relations, i.e., tables, as mentioned above. Each relation has a name and is formed from named

attributes or columns of data. Each tuple or row holds one value per attribute.

The greatest strength of the relational model is the simple logical structure that it forms. Behind this simple structure is a sophisticated theoretical foundation that is lacking in the first generation of DBMSs.

V: DATABASE SCHEMA

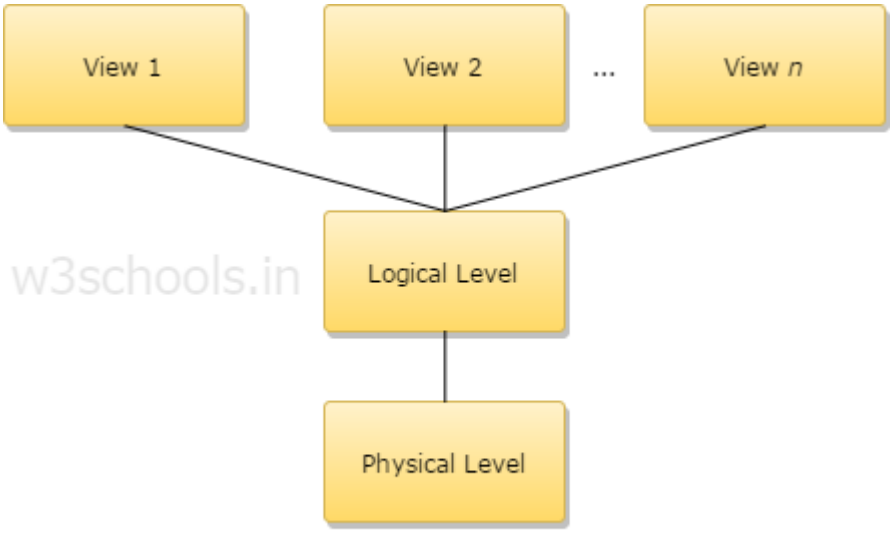
When you talk about the database, you must distinguish between the database schema, which is the logical blueprint of the database, and the database instance, which is a snapshot of the data in the database at a given instant in time. The concept of a relation corresponds to the programming language notion of a variable. In contrast, the concept of a relation schema corresponds to the programming languages' notion of the type definition. In other words, a database schema is a skeletal structure that represents the logical view of the complete database. It describes how the data is organized and how the relations among them are associated and formulates all the constraints that are to be applied to the data. In general, a relation schema consists of a directory of attributes and their corresponding domain.



Relational Model Terms

A schema can be defined as the design of a database. The overall description of the database is called the database schema. It can be categorized into three parts. These are:

- Physical Schema
- Logical Schema
- View Schema



Three Levels of Data Abstraction

A physical schema can be defined as the design of a database at its physical level. In this level, it is expressed how data is stored in blocks of storage.
A logical schema can be defined as the design of the database at its logical level. In this level, the programmers, as well as the database administrator (DBA), work. At this level, data can be described as certain types of data records that can be stored in the form of data structures. However, the internal details (such as an implementation of data structure) will be remaining hidden at this level.
View schema can be defined as the design of the database at the view level, which generally describes end-user interaction with database systems.

VI: DATA INDEPENDENCE

Logical Data Independence

Logical data independence can be defined as the immunity of the external schemas to changes in the conceptual schema.

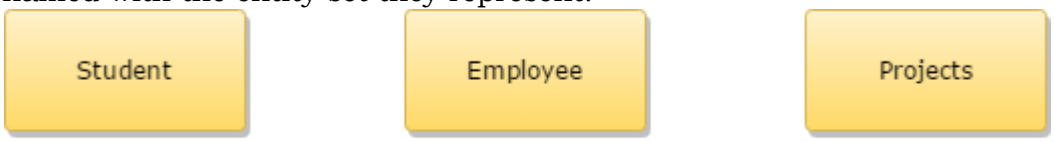
Physical data independence

Physical data independence can be defined as the immunity of the conceptual schema to changes in the internal schema.

VII: ENTITY RELATIONSHIP DIAGRAM

ER-Diagram is a pictorial representation of data that describes how data is communicated and related to each other. Any object, such as entities, attributes of an entity, sets of relationship, and other attributes of relationship, can be characterized with the help of the ER diagram.

Entities: They are represented using the rectangle-shaped box. These rectangles are named with the entity set they represent.



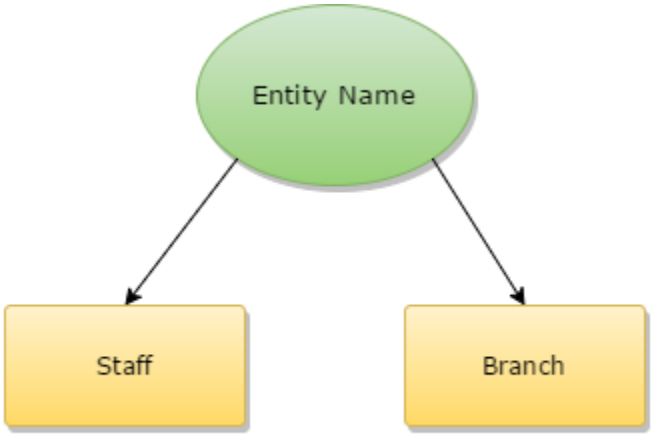
Entities

ER modeling is a top-down structure to database design that begins with identifying the important data called entities and relationships in combination with the data that must be characterized in the model. Then database model designers can add more details such as the information they want to hold about the entities and relationships, which are the attributes and any constraints on the entities, relationships, and attributes. ER modeling is an important technique for any database designer to master and forms the basis of the methodology.

- Entity type: It is a group of objects with the same properties that are identified by the enterprise as having an independent existence. The basic concept of the ER model is the entity type that is used to represent a group of 'objects' in the 'real world' with the same properties. An entity type has an independent existence within a database.
- Entity occurrence: A uniquely identifiable object of an entity type.

Diagrammatic Representation of Entity Types

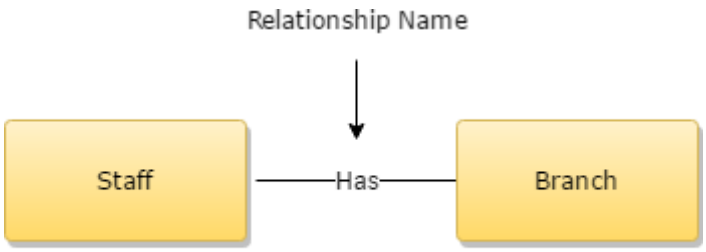
Each entity type is shown as a rectangle labeled with the name of the entity, which is usually a singular noun.



Diagrammatic representation of the Staff and Branch entity types.

What is Relationship Type?

A relationship type is a set of associations between one or more participating entity types. Each relationship type is given a name that describes its function.
Here is a diagram showing how relationships are formed in a database.



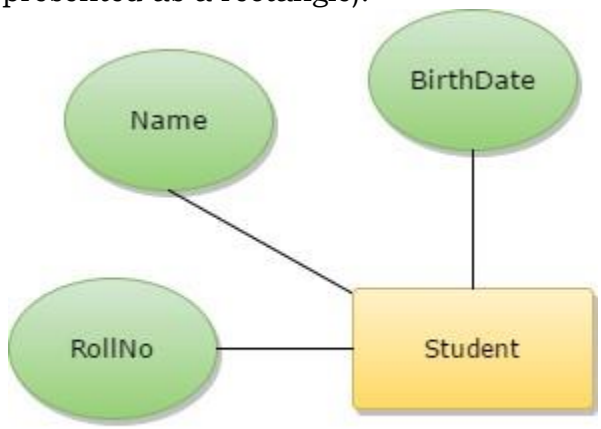
Branch has staff

What is a degree of Relationship?

The entities occupied in a particular relationship type are referred to as participants in that relationship. The number of participants involved in a relationship type is termed as the degree of that relationship.
In the above-figured example, "Branch has a staff", there is a relationship between two participating entities. A relationship of degree two is called binary degree (relationship).

What are Attributes?

Attributes are the properties of entities that are represented using ellipse-shaped figures. Every elliptical figure represents one attribute and is directly connected to its entity (which is represented as a rectangle).

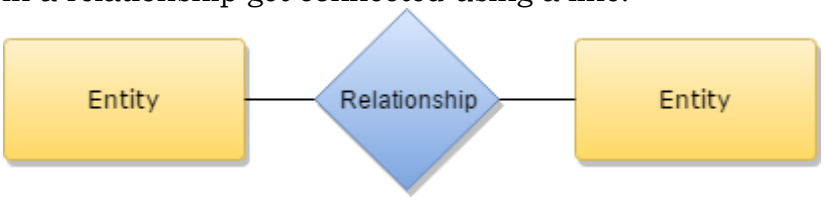


Attributes

It is to be noted that multi-valued attributes are represented using double ellipse like this:

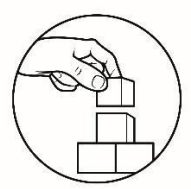
Relationships

A diamond-shaped box represents relationships. All the entities (rectangle-shaped) participating in a relationship get connected using a line.



There are four types of relationships. These are:

- One-to-one: When only a single instance of an entity is associated with the relationship, it is termed as '1:1'.
- One-to-many: When more than one instance of an entity is related and linked with a relationship, it is termed as '1:N'.
- Many-to-one: When more than one instance of an entity is linked with the relationship, it is termed as 'N:1'.
- Many-to-many: When more than one instance of an entity on the left and more than one instance of an entity on the right can be linked with the relationship, then it is termed as N:N relationship.



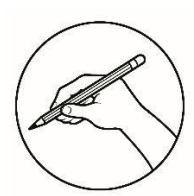
What's More

Database Management System (DBMS) is technology to create and manage databases and one of main program used for it is SQL Programming. SQL (Structured Query Language) is the standard database programming language to access and manipulate data in the database. One of its basic syntax are as follows:

```
SELECT *
FROM USERS;
```

Your task is to search on how the given SQL commands work, define each command line, its sample output and how does it relate to RDBMS.

TIP: You can use the reference from this link: https://www.w3schools.in/sql/



Assessment

Part I: Identification: Identify the defined terms and select the best answer on the box below:

DATA	INFORMATION	PROCESS
DATABASE	DBMS	USERS
DATABASE APPLICATION	DBMS SOFTWARE	DATABASE ENVIROMENT
RELATIONAL MODEL	RELATION	ATTRIBUTE
DOMAIN	TUPLE	ER-MODEL

1. These are facts concerning things such as people, objects or events.
2. Data that have been processed and presented in a form suitable for human interpretation, often with the purpose of revealing trends or patterns.
3. This involves acquisition, storage, manipulation, retrieval and distribution.
4. A shared collection of logically related data, designed to meet the information needs of multiple users in an organization.

5. Refers to the technology for creating and managing databases.
6. Main aim is to supply a way to store up and retrieve database information that is both convenient and efficient.
7. One component of DBMS, may be of any kind such as DB administrator, System developer, or database users.
8. One component of DBMS, may be Departmental, Personal, organization's and / or Internal.
9. One component of DBMS that allows users to create and manipulate database access.
10. A collective system of components that comprise and regulates the group of data, management, and use of data, which consist of software, hardware, people, techniques of handling database, and the data also.
11. Is the theoretical basis of relational databases, which is a technique or way of structuring data using relations, which are grid-like mathematical structures consisting of columns and rows.
12. In Relational Model, this is a table with columns and rows.
13. In Relational Model, this is a named column of a relation.
14. In Relational Model, this is the set of allowable values for attributes.
15. In Relational Model, this is a row of a relation.

Part II: Enumeration: Enumerate the following.

A. Types of Databases (3 pts).
B. Types of DBMS Architecture (3 pts).
C. Category of DBMS (3 pts).
D. Kinds of DBMS Data Independence (2 pts).
E. At least 3 types of Relationships in ER-Model (3 pts).

Part III: Draw the figure used for database modelling in Entity-Relationship Diagram.

A. Entity
B. Attribute
C. Relationship
D. One-to-one relationship
E. Many-to-many relationship

Assessment Part 1: 1 DATA 2 INFORMATION 3 PROCESS 4 DATABASE 5 DBMS 6 DBMS 7 USERS 8 DATABASE APPLICATION 9 DBMS SOFTWARE 10 DATABASE ENVIRONMENT 11 RELATIONAL MODEL 12 RELATION 13 ATTRIBUTE 14 DOMAIN 15 TUPLE	Assessment Part II: A: Bibliographic, Full-Text, Numeric, Images B: Centralized, Decentralized, Hierarchical C: Physical, Logical, View D: Logical, Physical E: 1-to-1, 1-to-many, many-to-one, many-to-many

References

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- *w3schools, Database Environment:* https://www.w3schools.in/dbms/database-environment/
- *w3schools, Database Architecture:* https://www.w3schools.in/dbms/database-architecture/
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- *w3schools, DBMS Data Schemas:* https://www.w3schools.in/dbms/data-schemas/
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